

R³AG 2025: The Second Workshop on Refined and Reliable Retrieval-Augmented Generation

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Abstract

In recent years, large language models (LLMs) have demonstrated remarkable capabilities across a broad spectrum of tasks, spanning language understanding, complex reasoning, and decision-making. However, they still face inherent limitations, such as *hallucinations* and *outdated parametric knowledge*. To mitigate these challenges, retrieval-augmented generation (RAG) has emerged as a promising technique and has attracted increasing attention. As RAG continues to be widely applied, an increasing number of challenges and limitations have surfaced, underscoring the urgent need for deeper, foundational research to advance and refine current RAG frameworks. Therefore, we propose to organize R³AG 2025, the second workshop on *Refined and Reliable Retrieval-Augmented Generation*, at SIGIR-AP 2025. This workshop seeks to bring together researchers and practitioners to re-examine and re-establish the core principles and practical implementations of refined and reliable RAG. This workshop will serve as a collaborative platform for academia and industry to exchange insights, discuss foundational issues and recent advancements. By the end of the workshop, we aim to arrive at a clearer understanding on promising directions for enhancing the reliability and applicability of RAG.

CCS Concepts

• Information systems → Information retrieval; • Computing methodologies → Natural language generation.

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Keywords

Retrieval-augmented Generation, Information Retrieval, Large Language Models, Reliability

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1 Background and Motivations

In recent years, large language models (LLMs) have made a transformative impact on artificial intelligence, revolutionizing a broad spectrum of fields, including natural language understanding, computer vision, and beyond. Despite these transformative impacts, LLMs still suffer from inherent limitations [10], such as hallucinations and outdated parametric knowledge. To address these issues, retrieval-augmented generation (RAG) has emerged as a promising direction, drawing increasing attention for its potential to enable dynamic integration of external information and knowledge. RAG has demonstrated its effectiveness across a wide range of applications [17], such as text-centric generation, code-related tasks and multimodal visual analysis. However, several challenges remain in achieving truly refined and reliable retrieval-augmented generation (R³AG). A typical RAG pipeline involves multiple key stages, including user intent comprehension, knowledge parsing, knowledge retrieval, and response generation. Each stage presents distinct challenges and plays a vital role in fulfilling user queries. For instance, open questions include: how to accurately understand user intent in long and complex dialogue contexts; how to parse rich, structured knowledge from documents containing tables and figures; how to ensure the reliability of retrieved knowledge; and how to refine the generated responses for faithfulness and coherence. This workshop aims to foster deeper investigation into these critical aspects of

R³AG and to bring together researchers dedicated to advancing the robustness and reliability of RAG systems.

Topic. The scope of this workshop covers key aspects of the RAG pipelines, including user intent comprehension, knowledge parsing, knowledge retrieval, and response generation. A detailed description of these topics can be found in Section 2.

Relevance. Generative large language models (LLMs) have become a prominent research focus within IR community. Their capabilities have significantly expanded the scope of IR and transformed the way users engage in information-seeking processes. Meanwhile, RAG stands out as one of the most impactful applications of IR in the context of LLMs, positioning IR as a crucial component in LLM-driven downstream tasks. Given this close interplay between IR and LLMs, R³AG 2025 is highly aligned with the goals and interests of the SIGIR-AP community.

Motivation. Recently, the potential of RAG has been demonstrated across a wide range of AI applications. However, fundamental research challenges remain in advancing current RAG methodologies. SIGIR-AP is a leading conference known for its strong focus on IR-related research and applications. We believe that organizing R³AG 2025 at SIGIR-AP 2025 is both timely and impactful. It can: (i) stimulate innovative research to address complex information-seeking demands; (ii) encourage the community to pursue in-depth investigations and practical applications in refined and reliable RAG; and (iii) further expand the visibility and influence of both SIGIR-AP and the R³AG workshop.

2 Fundamental Challenges and Topics

This year, we focus more on fundamental challenges in this field and anticipate rich, in-depth discussions during R³AG 2025.

User Intent Comprehension. Effective comprehension of user intent plays a crucial role in guiding both the retrieval process and the final response generation. However, understanding user intent can be particularly challenging, especially in the context of long or multi-turn dialogues. R³AG aims to assist users in clarifying their information needs throughout the interaction process. Relevant techniques include, but are not limited to, query expansion, query summarization [8], and query rewriting [13].

RAG on Complex Documents. RAG systems aim to deliver relevant knowledge chunks to enhance the generation capabilities of LLMs. However, existing methods often struggle to parse complex documents that contain embedded tables, figures, and other non-textual elements. Effectively extracting and utilizing tabular knowledge remains an open research challenge. Moreover, the strategy for dividing documents into chunks significantly impacts response quality—overly large chunks may introduce irrelevant information, while overly small chunks risk omitting essential content. Additionally, multi-hop document retrieval, which requires aggregating information from multiple sources or passages, poses another key challenge. These three issues—complex document parsing, chunking strategies, and multi-hop retrieval—warrant further investigation to advance the development of R³AG.

Reliable Retrieval for RAG. The inclusion of noisy or contradictory information during retrieval can significantly undermine the performance of existing RAG models. As a result, there is growing

interest in enhancing the robustness of RAG systems against harmful or counterfactual inputs, an aspect increasingly recognized as a critical performance indicator in recent studies [4, 15, 16]. Interestingly, recent research [7] shows that incorporating irrelevant documents can, in some cases, unexpectedly boost accuracy by over 30%, challenging the conventional assumption that such inclusion would necessarily degrade output quality. Motivated by these findings, R³AG aims to foster the development of specialized strategies for effective and resilient retrieval, while also highlighting the continued need for deeper investigation into the robustness and reliability of RAG systems.

Response Evaluation and Refinement. While RAG enhances LLMs by supplying additional context, challenges related to unreliable or inaccurate response generation persist. These issues may stem from incorrect information in the retrieved context [5] or inherent hallucination tendencies in the generation process [2]. Despite their significance, there remains a notable gap in understanding how these factors affect the overall quality of RAG outputs, as well as in developing effective strategies for mitigating them and refining model responses. To address this, R³AG is committed to the comprehensive evaluation and improvement of response quality in RAG-based LLMs across multiple dimensions, including relevance, faithfulness, negative response rejection, information integration, creative generation, and error correction.

Multimodal R³AG. Most current research focuses on textual RAG, despite the need for multimodality in many applications. Recent large-scale models like Flamingo [3], and GPT-4 [1] show significant multimodal capability when scaled to tens of billions of parameters and trained on extensive multimodal corpora. These large multimodal models require RAG even more than unimodal textual LLMs for external knowledge support. Therefore, this workshop also encourage discussions regarding R³AG for large multimodal models.

Reinforcement Learning for RAG. Inspired by the success of reinforcement learning (RL)-based models, such as OpenAI-o1 and DeepSeek-R1, recent studies [6, 9, 11, 12, 14] have begun exploring the use of RL to enhance RAG performance, for instance, by training policy models to more effectively search for relevant information. Despite the promising advancements, the effective integration of RL into RAG remains an open and challenging research problem.

3 Program Sketch

3.1 Workshop Format

The workshop is planned as a half-day event, featuring two invited talks and four oral research presentations. There are two encouraging types of invited talks: (i) Academic talks that focus on fundamental research concerning the adaptability and reliability of RAG; and (ii) Industrial talks that share practical insights on designing or applying refined and reliable RAG methods in real-world scenarios, such as information retrieval and generative systems. Each talk will be delivered in the form of a slide-based lecture, with a subsequent question-and-answer session.

3.2 Workshop Schedule

The tentative workshop schedule is outlined as follows:

- 9:00 - 9:10 Welcome & opening

- 9:10 - 9:50 Academic invited talk
- 9:50 - 10:30 Industrial invited talk
- 10:30 - 10:50 Coffee break
- 10:50 - 11:40 Oral paper talks
- 11:40 - 12:00 Panel discussion

Tentative Speakers. The tentative speakers include academia researchers, such as Prof. Xiangyu Zhao from the City University of Hong Kong, and Prof. Wenqi Fan from The Hong Kong Polytechnic University and industrial staff, such as Dr. Alexandros Karatzoglou from Amazon.

Panel Discussion. We are also considering concluding the workshop with a panel discussion. The inclusion of this session will depend on the availability of suitable panelists attending SIGIR-AP and the number of accepted research papers.

Contingency Plan. To ensure the success of the workshop, we have developed a contingency plan that outlines key requirements (e.g., the venue, speakers, and materials) and assesses potential risks. Backup venues and alternate speakers will be arranged in advance, and a dedicated support team will be available on-site. A designated lead will oversee the communication strategy, while clearly assigned roles will enable last-minute organizers to efficiently manage registration, technical support, and logistical coordination.

3.3 Selection Process

Each invited speaker will be a highly respected member of the research community, and All invitations will be issued with the unanimous agreement of the organizers. Paper submissions to the workshop will follow a standard peer-review process. We anticipate receiving approximately 6–10 submissions and expect to accept 3–4 papers. Each submission will be reviewed by at least two members of the program committee, with final decisions made by the senior PC members or workshop organizers. Authors will receive detailed reviewer feedback along with a formal notification letter.

Tentative Program Committee. In addition to the current organizers, we also plan to invite the following tentative PC members: (1) Prof. Chao Huang from University of Hong Kong, (2) Prof. Xiangyu Zhao from the City University of Hong Kong, (3) Prof. Andrew Yates from University of Amsterdam, and (4) Dr. Alexandros Karatzoglou from Amazon.

3.4 Online Materials

A dedicated website for R³AG 2025, along with access to the previous R³AG 2024, will be made available online. All relevant materials, including talk information, presentation slides, referenced papers, speaker details, and related open-source projects, will be accessible through this site.

3.5 Workshop Advertisement

R³AG 2025 will be promoted across various social media platforms to enhance visibility and encourage paper submissions. These platforms include, but are not limited to, Twitter, Facebook, and WeChat. In addition, the organizers will circulate personalized email invitations to further publicize the workshop.

4 Related Workshops

A number of highly related workshops can be listed as follows: (1) Information Retrieval’s Role in RAG Systems¹(SIGIR 2025), (2) Robust Information Retrieval²(SIGIR 2025), (3) Multimodal Augmented Generation via Multimodal Retrieval³(ACL 2025), (4) Evaluation of RAG Systems⁴(ECIR 2025), (5) AI Agent for Information Retrieval⁵(KDD 2025).

The Information Retrieval’s Role in RAG Systems workshop at SIGIR 2025 continues to explore retrieval’s foundational role in the design and performance of RAG frameworks. Complementing this, the Robust Information Retrieval workshop at SIGIR 2025 focuses on the reliability of retrieval methods. At ACL 2025, the Multimodal Augmented Generation via Multimodal Retrieval workshop highlights the growing importance of multimodal inputs in RAG. The Evaluation of RAG Systems workshop at ECIR 2025 focuses on establishing rigorous benchmarks and evaluation protocols for RAG, emphasizing the need for standardized assessments. Meanwhile, the AI Agent for Information Retrieval workshop at KDD 2025 investigates how autonomous agents interact with retrieval modules in complex environments, offering valuable insights for building adaptive and context-aware RAG systems.

R³AG 2025 builds upon these efforts by uniquely centering on refined and reliable RAG techniques. As a complementary venue, it aims to bring together academic and industrial communities through invited talks and research presentations.

5 Organizers and Attendance Probabilities

Dr. Haitao Yu (99%) is a Tenured Associate Professor at University of Tsukuba, Japan, leading the *Information Intelligence* research group. His research currently focuses *Retrieval-augmented LLM*, *Trustworthy Information Retrieval* and *Commonsense Reasoning*. He has published a series of papers on top international conferences (e.g., WSDM, CIKM, SIGIR, WWW, ECIR, and AAAI) and journals (e.g., Information Processing and Management, and Information Retrieval Journal). He has played an active role in co-organizing multiple conferences (SIGIR-AP 2024, Australasian Database Conference 2024) and workshops (R³AG at SIGIR-AP 2024, EReL@MIR at The Web Conference 2025, AISTORY at ICCV 2025).

Yubo Fang (99%) is a PhD student of the Information Intelligence research group at the University of Tsukuba, Japan. His currently research focuses on Large Language Model and Retrieval-Augmented Generation.

Dr. Xuri Ge (99%) is an Assistant Professor at the School of Artificial Intelligence, Shandong University (SDU), China. He has published over 20 papers in prestigious conferences and journals, including ICLR, ACM MM, SIGIR, NeurIPS, IP&M, TKDE, CIKM and ACM TIST, etc. His current research focuses on information retrieval, efficient multimodal representation learning, and multimodal large language models. Xuri has organized EReL@MIR Workshop in WWW and NIP@IR workshop in SIGIR2025.

¹<https://coda.io/@rstless-group/ir-rag-sigir25>

²<https://sigir-2025-workshop-on-robust-ir.github.io>

³<https://nlp.jhu.edu/magmar/>

⁴<https://eval4rag.github.io>

⁵<https://sites.google.com/view/ai4ir/kdd-2025>

Dr. Xin Xin (90%) is a Tenure-Track Associate Professor at the School of Computer Science and Technology of Shandong University, China. Before that, he earned his Ph.D. degree from the University of Glasgow. His current research interests include information retrieval, natural language processing, and reinforcement learning. He has published more than 40 papers in top conferences (e.g., WWW, SIGIR, ACL, WSDM) and journals (e.g., TOIS, TKDE), and received the Best Paper Honor Mention at WSDM 2024. He has organized the DRL4IR workshop in SIGIR and KEIR workshop in ECIR.

Zihan Wang (80%) is a fourth-year PhD student at the University of Amsterdam (UvA). He has published over ten papers in prestigious conferences, including KDD, SIGIR, CCS, and WSDM. He received the Best Student Paper Award at WSDM 2018. His current research focuses on information extraction, knowledge graph embedding, and the reliability of large language models.

Junchen Fu (80%) is a PhD student at University of Glasgow. He has contributed to several leading AI conferences and journals, including SIGIR, WWW, WSDM and TPAMI. He also serves as the PC member and reviewer for top tier conferences and journals, such as ICLR, SIGIR, SIGKDD, WWW, etc. His research primarily focuses on enhancing the efficiency of adapting multimodal foundation models for recommender systems. He has organized EReL@MIR Workshop in WWW2025.

Prof. Joemon M. Jose (60%) is a Professor at the School of Computing Science, University of Glasgow, Scotland. His research focuses on the following themes: (i) Recommendation Systems; (ii) Facial Expression Analysis; (iii) Detection of Conflict in Social Media; (iv) Legal Outcome Prediction and (v) Automated Judgment Generation. He has published over 350 papers with more than 11,000 Google Scholar citations, and an h-index of 52. He leads the GAIR-lab (<https://gair-lab.github.io/>) investigating research issues related to the above themes. He has been serving as the program committee chair and member for numerous top international conferences (e.g., SIGIR, WWW, and ECIR). He was a PC co-chair for SIGIR-AP 2024.

Dr. Weizhi Ma (90%) is a Research Assistant Professor at the Institute for AI Industry Research (AIR) at Tsinghua University, China. He earned his B.E. and Ph.D. in the Department of Computer Science and Technology from Tsinghua University. Dr. Ma's research focuses on information retrieval, recommender systems, and LLM-powered agents. He has published over 70 papers in leading international conferences and journals, including TOIS, TKDE, SIGIR, KDD, AAAI, IJCAI, WSDM, CIKM, etc. His accolades include the Best Paper Honorable Mention Award at SIGIR 2020 and several other paper awards. In addition to his research, Dr. Ma is the Secretary of the Youth Working Committee of the Chinese Information Processing Society of China and serves as an Assistant Editor for ACM TOIS. He is a recipient of the Young Elite Scientists Sponsorship Program by CAST and the Shuimu Tsinghua Scholar Program.

Dr. Zhaochun Ren (80%) is an Associate Professor at Leiden University, Netherlands. His research interests focus on joint research in IR and natural language processing, with an emphasis on conversational information seeking, question-answering, and recommender systems. He aims to develop intelligent systems that can

address complex user requests and solve core challenges in both information retrieval and natural language processing towards that goal. In addition to his academic experience, he worked on e-commerce search and recommendation at JD.com for 2+ years. He has co-organized workshops at SIGIR (2020), WSDM (2019, 2020), and ECIR 2025.

References

- [1] Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altschmidt, Sam Altman, Shyamal Anadkat, et al. 2023. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774* (2023).
- [2] Vaibhav Adlakha, Parishad BehnamGhader, Xing Han Lu, Nicholas Meade, and Siva Reddy. 2024. Evaluating Correctness and Faithfulness of Instruction-Following Models for Question Answering. *Trans. Assoc. Comput. Linguistics* 12 (2024), 681–699.
- [3] Jean-Baptiste Alayrac, Jeff Donahue, Pauline Luc, Antoine Miech, Iain Barr, Yana Hasson, Karel Lenc, Arthur Mensch, Katherine Millican, Malcolm Reynolds, Roman Ring, Eliza Rutherford, Serkan Cabi, Tengda Han, Zhitao Gong, Sina Samangooei, Marianne Monteiro, Jacob L. Menick, Sebastian Borgeaud, Andy Brock, Aida Nematzadeh, Sahand Sharifzadeh, Mikolaj Binkowski, Ricardo Barreira, Oriol Vinyals, Andrew Zisserman, and Karén Simonyan. 2022. Flamingo: a Visual Language Model for Few-Shot Learning. In *NeurIPS*.
- [4] Jinheon Baek, Soyeong Jeong, Minki Kang, Jong C. Park, and Sung Ju Hwang. 2023. Knowledge-Augmented Language Model Verification. In *EMNLP*. 1720–1736.
- [5] Ning Bian, Hongyu Lin, Peilin Liu, Yaojie Lu, Chunkang Zhang, Ben He, Xianpei Han, and Le Sun. 2023. Influence of external information on large language models mirrors social cognitive patterns. *arXiv preprint arXiv:2305.04812* (2023).
- [6] Mingyang Chen, Tianpeng Li, Haoze Sun, Yijie Zhou, Chenzheng Zhu, Haofen Wang, Jeff Z. Pan, Wen Zhang, Huajun Chen, Fan Yang, Zenan Zhou, and Weipeng Chen. 2025. ReSearch: Learning to Reason with Search for LLMs via Reinforcement Learning. *arXiv:2503.19470 [cs.AI]* <https://arxiv.org/abs/2503.19470>
- [7] Florin Cuceranu, Giovanni Trappolini, Federico Siciliano, Simone Filice, Cesare Campagnano, Yoelle Maarek, Nicola Tonellotto, and Fabrizio Silvestri. 2024. The power of noise: Redefining retrieval for rag systems. *arXiv preprint arXiv:2401.14887* (2024).
- [8] Darren Edge, Ha Trinh, Newman Cheng, Joshua Bradley, Alex Chao, Apurva Mody, Steven Truitt, and Jonathan Larson. 2024. From local to global: A graph rag approach to query-focused summarization. *arXiv preprint arXiv:2404.16130* (2024).
- [9] Jingsheng Gao, Linxu Li, Ke Ji, Weiyan Li, Yixin Lian, yuzhuo fu, and Bin Dai. 2025. SmartRAG: Jointly Learn RAG-Related Tasks From the Environment Feedback. In *The Thirteenth International Conference on Learning Representations*. <https://openreview.net/forum?id=OCd3cfullp>
- [10] Lei Huang, Weijiang Yu, Weitao Ma, Weihong Zhong, Zhangyin Feng, Haotian Wang, Qianglong Chen, Weihua Peng, Xiaocheng Feng, Bing Qin, and Ting Liu. 2025. A Survey on Hallucination in Large Language Models: Principles, Taxonomy, Challenges, and Open Questions. *ACM Trans. Inf. Syst.* 43, 2, Article 42 (Jan. 2025), 55 pages. doi:10.1145/3703155
- [11] Yingqian Min Jie Chen Zhipeng Chen Wayne Xin Zhao Ji-Rong Wen Yang Lu Xu Miu Huatong Song, Jinhao Jiang. 2025. R1-Searcher: Incentivizing the Search Capability in LLMs via Reinforcement Learning. (2025). <https://github.com/RUCAIBox/R1-searcher>
- [12] Bowen Jin, Hansi Zeng, Zhenrui Yue, Jinsung Yoon, Serkan Arik, Dong Wang, Hamed Zamani, and Jiawei Han. 2025. Search-r1: Training llms to reason and leverage search engines with reinforcement learning. *arXiv preprint arXiv:2503.09516* (2025).
- [13] Shengyu Mao, Yong Jiang, Boli Chen, Xiao Li, Peng Wang, Xinyu Wang, Pengjun Xie, Fei Huang, Huajun Chen, and Ningyu Zhang. 2024. RaFe: Ranking Feedback Improves Query Rewriting for RAG. *arXiv preprint arXiv:2405.14431* (2024).
- [14] Hao Sun, Zile Qiao, Jiayan Guo, Xuanbo Fan, Yingyan Hou, Yong Jiang, Pengjun Xie, Fei Huang, and Yan Zhang. 2025. ZeroSearch: Incentivize the Search Capability of LLMs without Searching. *arXiv preprint arXiv:2505.04588* (2025).
- [15] Ori Yoran, Tomer Wolfson, Ori Ram, and Jonathan Berant. 2023. Making retrieval-augmented language models robust to irrelevant context. *arXiv preprint arXiv:2310.01558* (2023).
- [16] Wenhao Yu, Hongming Zhang, Xiaoman Pan, Kaixin Ma, Hongwei Wang, and Dong Yu. 2023. Chain-of-note: Enhancing robustness in retrieval-augmented language models. *arXiv preprint arXiv:2311.09210* (2023).
- [17] Penghao Zhao, Hailin Zhang, Qinhao Yu, Zhengren Wang, Yunteng Geng, Fangcheng Fu, Ling Yang, Wentao Zhang, and Bin Cui. 2024. Retrieval-Augmented Generation for AI-Generated Content: A Survey. *arXiv preprint arXiv:2402.19473* (2024).